RESP-100 · BASIC EKG INTERPRETATION · WEEK 6

Sinus Rhythms

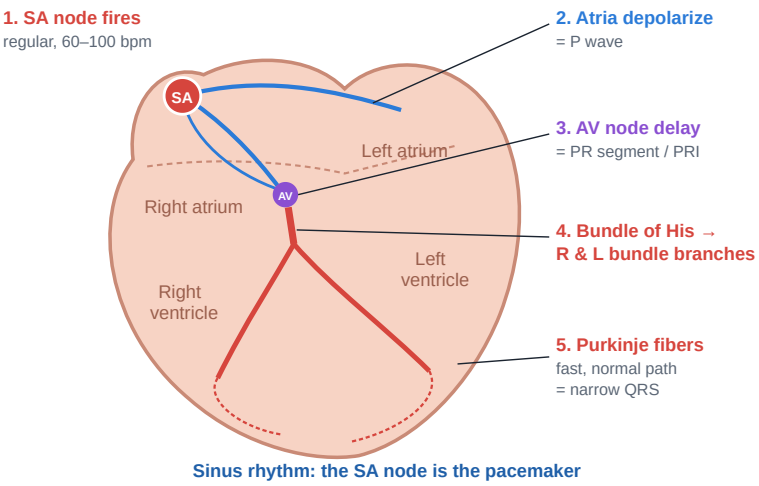
Walraven, *Basic Arrhythmias* 8th ed. · Chapter 4 · Lecture by Michael V Hyland (55 min)

In one line: four rhythms all start in the SA node. They look normal beat by beat. The only things that change are **how fast** (rate) and **how evenly spaced** (regularity) the beats are.

1 · What "sinus" means | 2 · Your measuring toolkit | 3 · Normal sinus rhythm | 4 · Sinus bradycardia
5 · Sinus tachycardia | 6 · Sinus arrhythmia | Cheat sheet | Practice quiz

1 What makes a rhythm "sinus"?

Big idea: In every sinus rhythm the **SA node is the pacemaker** and each impulse travels the **normal pathway** (AV node → bundle of His → bundle branches → Purkinje). So every sinus beat has an upright, rounded P wave, a normal PRI and a narrow QRS.



The normal conduction pathway. The colours match the waves they make on the ECG.

What "sinus P waves" look like

- **Upright** (positively deflected)
- **Uniform**: they all look the same. If you laid one over another, they'd match.
- Usually a nice **rounded dome**, not pointed, flat or notched
- One P in front of every QRS

"Conducted normally" = narrow QRS

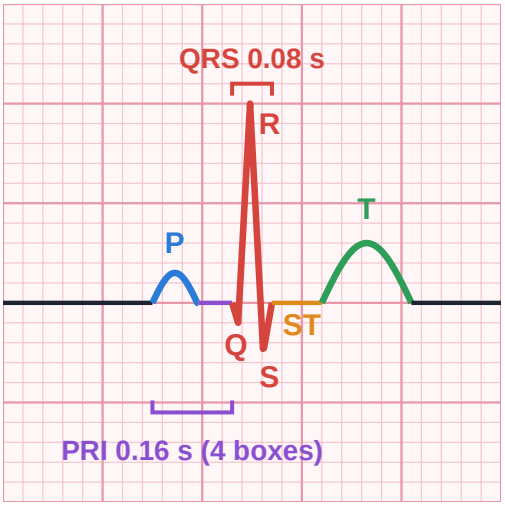
- The impulse uses the fast nerve pathway, so the ventricles depolarize quickly
- Normal = narrow = **0.12 s or less (3 small boxes or less)**
- The instructor uses "normal" and "narrow" interchangeably

WHAT THE INSTRUCTOR STRESSED (0:00:52)

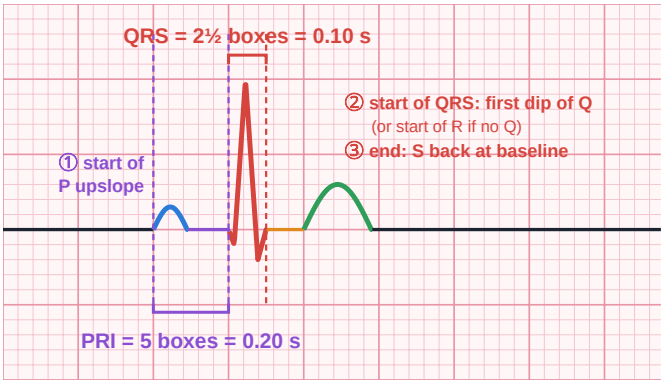
The homework has 10 strips, and every one of them is one of today's 4 sinus rhythms. Each strip has **6 blanks worth 1 point each**: regularity, heart rate, P waves, PRI, QRS and interpretation.

2 Your measuring toolkit

Big idea: First gather the clues (regularity → rate → P waves → PRI → QRS). Then compare them with each rhythm's rules and see where they fit best. That's how you interpret a strip.

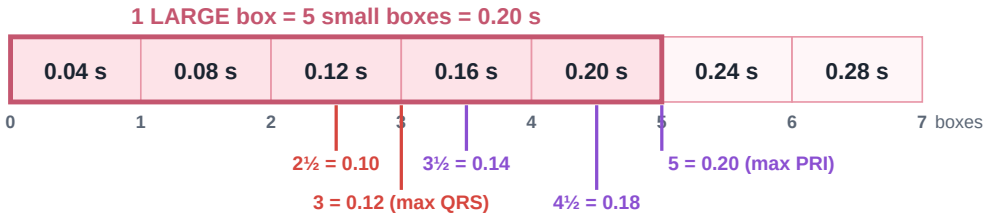


Anatomy of one sinus beat (1 small box = 0.04 s).



Where to start and stop when you measure the PRI and the QRS.

Convert boxes to seconds (the final exam asks for time, not boxes)



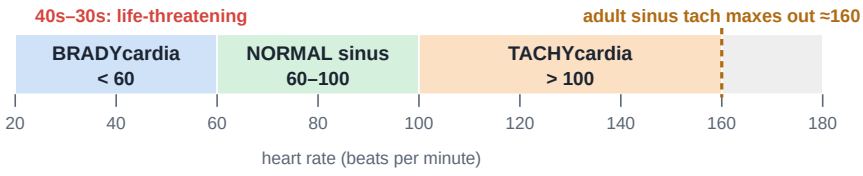
Multiply the number of small boxes by 0.04. These are the values from this lecture's strips.

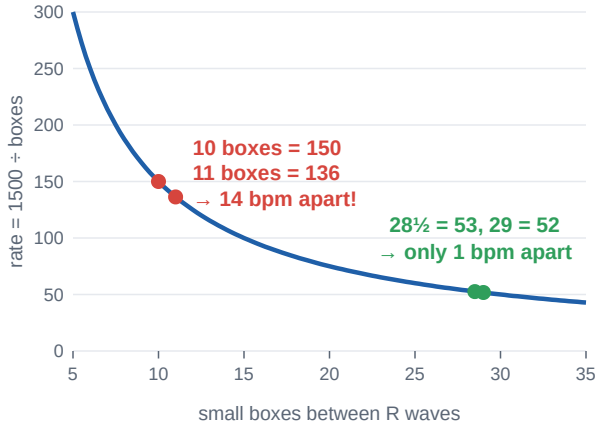
Step 1 · Regular or irregular?

- Count the **small boxes between R waves** all the way across the strip
- If they're all **within 1½ small boxes** of each other, the rhythm is **regular**
- A bigger difference than that means it's irregular

Step 2 · Heart rate

- **Regular:** 1500 ÷ small boxes (17 boxes → 88)
- **Irregular:** count the R waves on the 6-second strip × 10
- Slower rate = R waves farther apart; faster rate = closer together





Why counting carefully matters more at FAST rates.

INSTRUCTOR, 0:21:50

At a slow rate, being off by ½ or 1 box barely changes the answer. Make the same mistake on a rhythm above 100 and you could be **20–30 beats off**. Get comfortable counting small boxes.

GRADING GRACE

Your answer counts as right if it is **within one small box** of his. Measuring is "not an exact science" (on a 12-lead the machine measures it for you).

PAPER TRICK

Lay a piece of paper on the strip, mark where the wave starts and where it ends, then slide the paper down into a clear row of boxes to count.

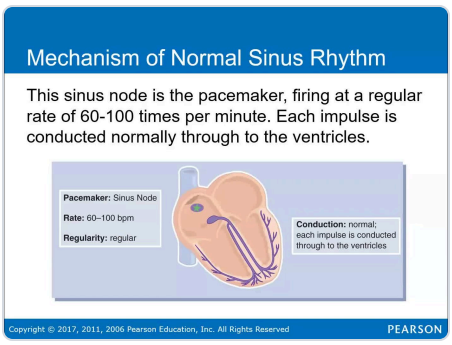
MEASURE MORE THAN ONE BEAT

Check the PRI on several beats to make sure it's **constant**. Later rhythms have PRIs that change. Skip beats with a slanted, fuzzy P upslope and use the clearly defined ones.

3 Normal Sinus Rhythm (NSR)

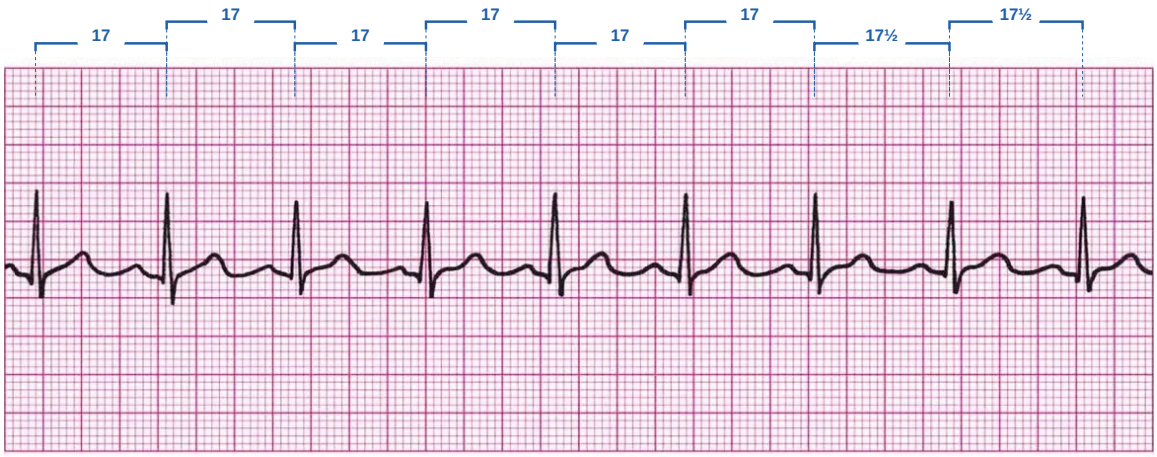
Big idea: When **everything is normal** (regular rhythm, rate 60–100, sinus P waves, normal PRI, narrow QRS), the strip can only be NSR.

Rules for NSR	
Regularity	Regular
Rate	60–100 bpm
P wave	Upright, uniform; one before every QRS
PRI	0.12–0.20 s and constant (3–5 boxes)
QRS	Less than 0.12 s (narrow)



Textbook slide: the mechanism of NSR.

Worked strip from class (0:05:35)



The NSR strip from the lecture, with the instructor's R-to-R counts in small boxes.

Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (17–17½)	1500 ÷ 17 = **88**	Upright, uniform, rounded	5 boxes = **0.20 s** (some 4½ = 0.18)	2½ boxes = **0.10 s**	**Normal sinus rhythm**

HOW HE MEASURED THE QRS

The Q wave starts right on a dark line, and the S wave ends halfway between two small-box lines. That's 1, 2, and a half = 2½ boxes = 0.08 + 0.02 = **0.10 s**.

TEST FORMAT (0:16:52)

The next test has **15 strips like the homework** (sinus, atrial and junctional rhythms). The **final is multiple choice** and asks for PRI and QRS **in seconds**, so practise converting.

4 Sinus Bradycardia

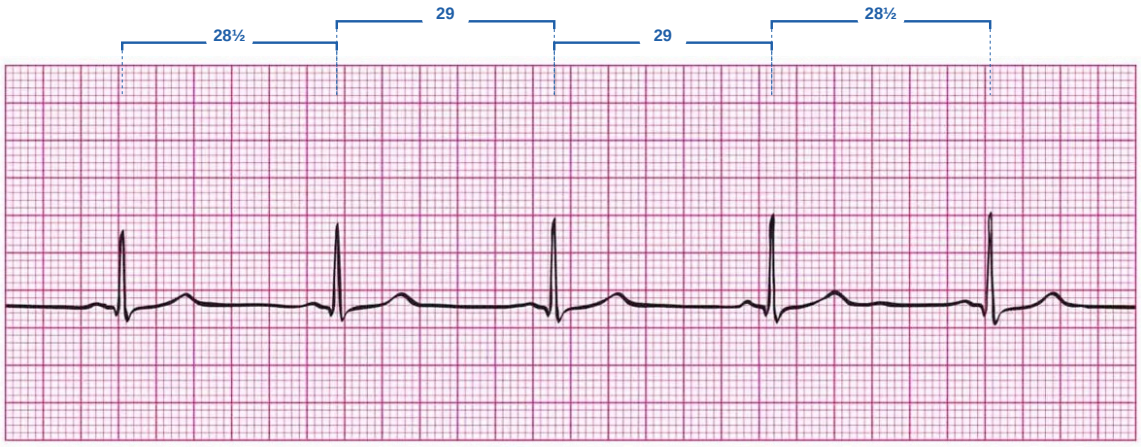
Big idea: This is NSR in every way **except the rate is below 60**. The SA node still fires regularly and conduction is normal. The R waves are just farther apart, because you divide a bigger number of boxes into 1500.

Rules for sinus bradycardia	
Regularity	Regular
Rate	**Less than 60 bpm**
P wave	Upright, uniform; one before every QRS
PRI	0.12–0.20 s and constant
QRS	Less than 0.12 s



A drawn example: R waves 29 small boxes apart.

Worked strip from class (0:19:26)



The sinus bradycardia strip from the lecture.

Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (28½–29)	1500 ÷ 29 ≈ **51–52**	Upright, uniform	3½ boxes = **0.14 s**	≈3 boxes = **0.12 s** (measured to the return to the isoelectric line)	**Sinus bradycardia**

Clinical picture: why a "normal-looking" rhythm can still be dangerous



So: check for a palpable pulse and measure BP. HR in the 40s or 30s = definitely life-threatening.

An ECG shows electrical activity, not mechanical activity. Always assess the patient too.

WHEN HR 52 IS OK

It can be normal during **sleep**, and in very fit **athletes** (marathon runners rest at about 50). It's **not** normal for someone up and active whose usual rate is 80–90.

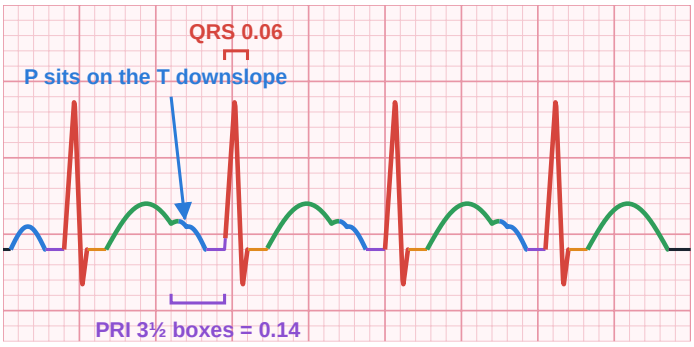
LIFE-THREATENING

A rate that drops into the **40s or 30s** is definitely life-threatening. "Pick up the phone and dial 911," especially if their BP is dropping.

5 Sinus Tachycardia

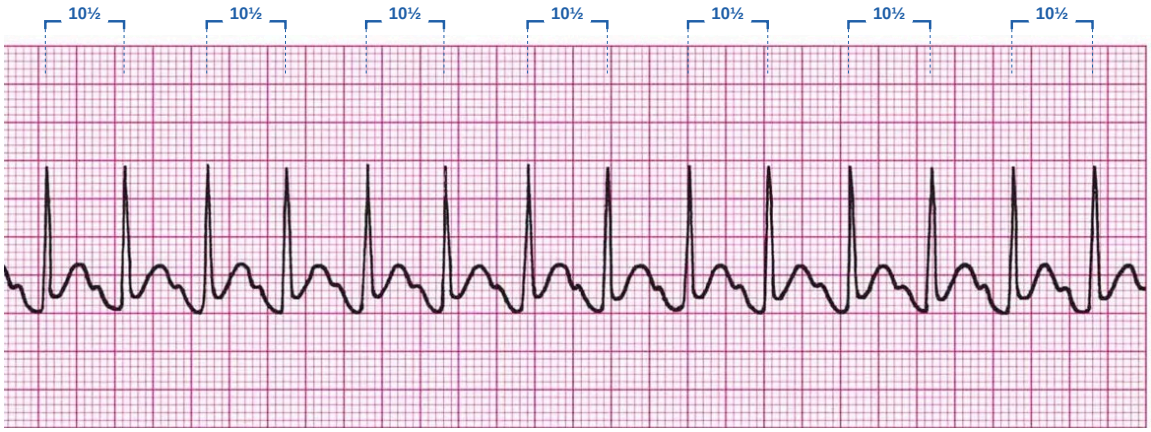
Big idea: This is NSR in every way **except the rate is above 100**. The R waves crowd together, so the **P wave can hide on the T wave**.

Rules for sinus tachycardia	
Regularity	Regular
Rate	**Greater than 100 bpm** (adults: usually not over 160)
P wave	Upright, uniform; one before every QRS. It may be only partly visible at high rates
PRI	0.12–0.20 s and constant
QRS	Less than 0.12 s



At about 140 bpm the P wave starts on the downslope of the previous T wave. There's no Q wave here, so the PRI ends at the start of the R wave.

Worked strip from class (0:35:27)



The sinus tachycardia strip from the lecture. Most spacings are 10½ boxes; a few are close to 10.

Regularity	Rate	P waves	PRI	QRS	Interpretation
Regular (10½)	1500 ÷ 10.5 = **143**	Upright, uniform (the "notch" on the T downslope)	3½ boxes = **0.14 s**	1½ boxes = **0.06 s**	**Sinus tachycardia**

INSTRUCTOR, 0:34:14

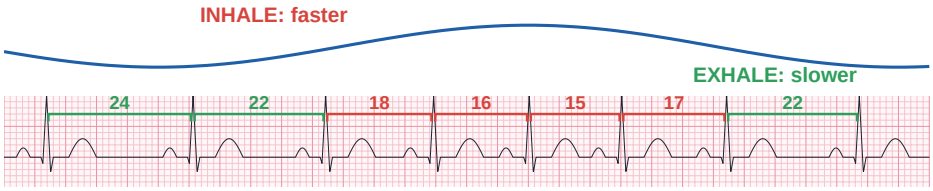
In adults, sinus tach **maxes out at about 160** (it can go higher in kids). You'll need this fact later in the course.

NOT EVERY QRS HAS A Q, R AND S

Just find where the QRS **begins** and where it **ends**, and count the boxes. If there's no Q wave, measure the PRI to the start of the R wave.

6 Sinus Arrhythmia

Big idea: "A-" means **without**, so arrhythmia means "without rhythm", i.e. **uneven spacing**. The SA node is still the pacemaker, but its rate rises and falls **with breathing**. Everything else is normal.



Breathe in and the rate speeds up (R waves get closer). Breathe out and it slows (R waves spread out). The numbers are R-to-R spacings in small boxes.

Rules for sinus arrhythmia	
Regularity	**Irregular**
Rate	Usually 60–100 bpm
P wave	Upright, uniform; one before every QRS
PRI	0.12–0.20 s and constant
QRS	Less than 0.12 s

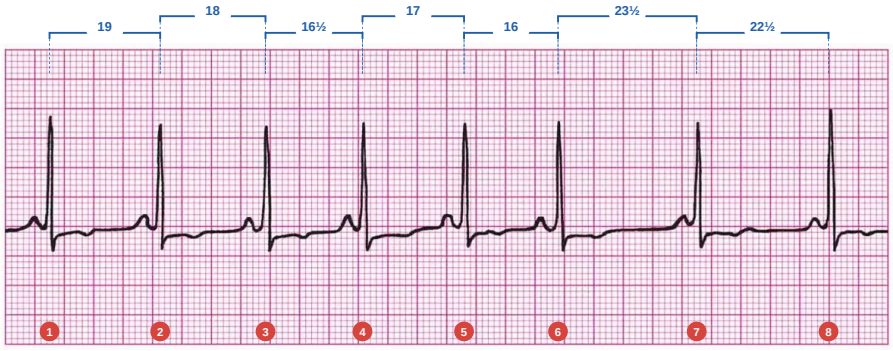


Irregular? Don't use 1500 ÷ boxes. Count the R waves in 6 seconds (tick marks 3 s apart) × 10. Here 8 × 10 = 80.

INSTRUCTOR'S "SNEAKY" COMBOS (0:44:33)

Two problems can happen together. Irregular **and** under 60 = **sinus brady-arrhythmia**. Irregular **and** over 100 = sinus tachy-arrhythmia. "I've been known to sneak a couple things in," but he said he **won't give one above 100**.

Worked strip from class (0:45:53)

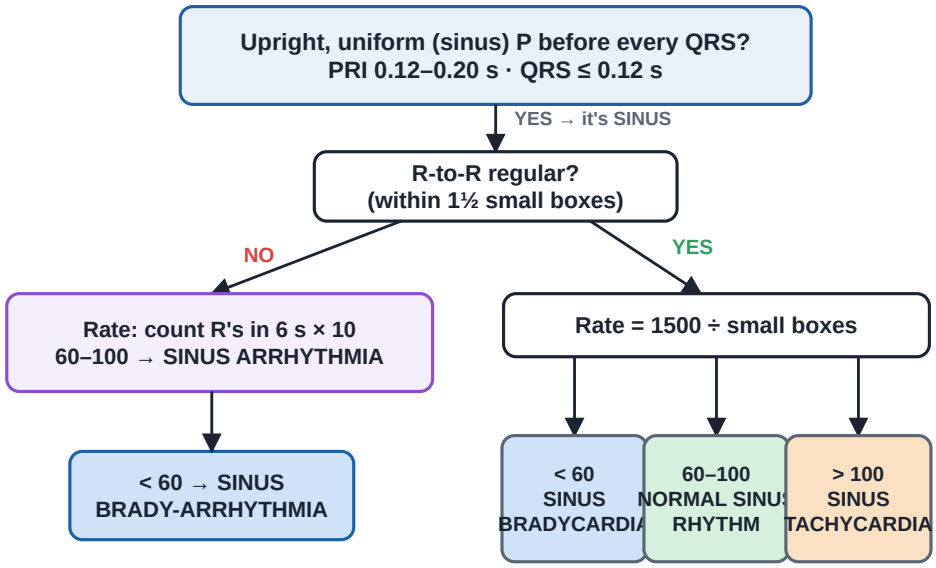


The R-to-R spacings vary from 16 to 23½ boxes, far more than 1½ boxes apart, so the rhythm is irregular. 8 R waves on the 6-second strip × 10 = 80.

Regularity	Rate	P waves	PRI	QRS	Interpretation
Irregular (16–23½)	8 R's × 10 = **80**	Upright, uniform	3½ boxes = **0.14 s**	2½ boxes = **0.10 s**	**Sinus arrhythmia**

★ Cheat sheet: the 4 sinus rhythms

	Normal sinus rhythm	Sinus bradycardia	Sinus tachycardia	Sinus arrhythmia
Regularity	Regular	Regular	Regular	**Irregular** (>1½ boxes)
Rate	60–100	**< 60**	**> 100** (adult max ≈160)	usually 60–100
P waves	Upright, uniform, rounded; one before every QRS (in tachycardia it may sit on the T wave)			
PRI	0.12–0.20 s (3–5 small boxes) and constant			
QRS	Narrow: 0.12 s or less (3 small boxes or less)			
Rate method	1500 ÷ boxes	1500 ÷ boxes	1500 ÷ boxes (count carefully!)	R's in 6 s × 10
Class strip	88 · 0.20 · 0.10	52 · 0.14 · 0.12	143 · 0.14 · 0.06	80 · 0.14 · 0.10



Decision tree for naming a sinus rhythm.



NSR: 17 boxes → 88 bpm



Sinus brady: 29 boxes → 52 bpm



Sinus tach: 10½ boxes → 143 bpm



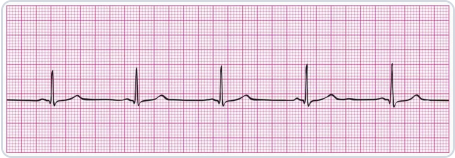
Sinus arrhythmia: uneven spacing

MEMORY HOOKS

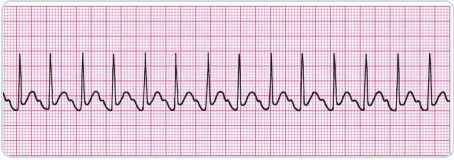
Brady = slow (think "brake"). **Tachy** = fast (think "tachometer"). **A-rrhythmia** = "without rhythm" = uneven spacing. The rhythm's **first name is always "sinus"** because the SA node is the pacemaker in all four.

? Practice quiz

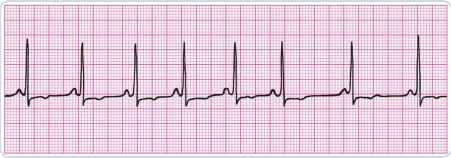
1. What is the pacemaker in all four sinus rhythms, and what does "conducted normally" tell you about the QRS?
2. Convert to seconds: (a) 2½ small boxes (b) 3½ small boxes (c) 5 small boxes.
3. The R-to-R spacings on a strip are 17, 17, 17½ and 17 boxes. Is it regular? What's the rate?
4. Why does a 1-box counting error matter more at a rate of 140 than at 52?
5. How do you calculate the heart rate for an irregular rhythm?
6. Identify strips A, B and C below. Fill in all 6 blanks for each (regularity, rate, P waves, PRI, QRS, interpretation).
7. On strip B, where are the P waves hiding?
8. On strip C, what is the giveaway that rules out NSR?
9. What happens to the heart rate during inspiration in sinus arrhythmia? During expiration?
10. A strip is irregular (because of breathing) with a rate of 50 and normal P, PRI and QRS. Name it.
11. Adult sinus tachycardia usually doesn't go above what rate?
12. A patient in sinus bradycardia at 45 is confused and sweaty. Use the cardiac output formula to explain why.
13. Where does the PRI end when the QRS has no Q wave?



Strip A



Strip B



Strip C

Answers. 1) The SA node. The impulse uses the normal pathway, so the QRS is narrow (0.12 s or less). 2) (a) 0.10 s (b) 0.14 s (c) 0.20 s. 3) Regular (all within 1½ boxes); 1500 ÷ 17 = 88. 4) 1500 ÷ boxes changes much faster when the box count is small: 10 → 150 but 11 → 136 (14 bpm apart), while 28½ → 53 and 29 → 52. 5) Count the R waves on the 6-second strip and multiply by 10. 6) A: regular (28½–29), rate ≈ 52, P upright and uniform, PRI 0.14, QRS ≈ 0.12 → **sinus bradycardia**. B: regular (10½), rate 143, P upright and uniform, PRI 0.14, QRS 0.06 → **sinus tachycardia**. 7) On the downslope of each T wave (the small notch). 8) The R-to-R spacing is irregular (more than 1½ boxes apart). C: irregular (16–23½ boxes) with normal sinus P waves, PRI 0.14, QRS 0.10, rate 80 → **sinus arrhythmia**. 9) Inspiration: the rate speeds up (R waves closer). Expiration: it slows (R waves farther apart). 10) **Sinus brady-arrhythmia.** 11) About 160 bpm. 12) CO = SV × HR. The low HR drops CO, which drops BP, so the brain and skin are under-perfused (confusion, agitation, diaphoresis). Rates in the 40s–30s are life-threatening. 13) At the beginning of the R wave (the first deflection of the QRS).